



A web platform for AI-powered generation, viewing, and local editing of 3D models

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Main screen: 3D viewer and control panel

About the project

Text-based AI generators create an entire model at once, while professional editors require 3D skills. 3D Forge sits between them: you open a model in the browser, click on the part you want, describe the edit in words — the system assembles a "spatial prompt" (text + coordinates + region) and applies a local change.

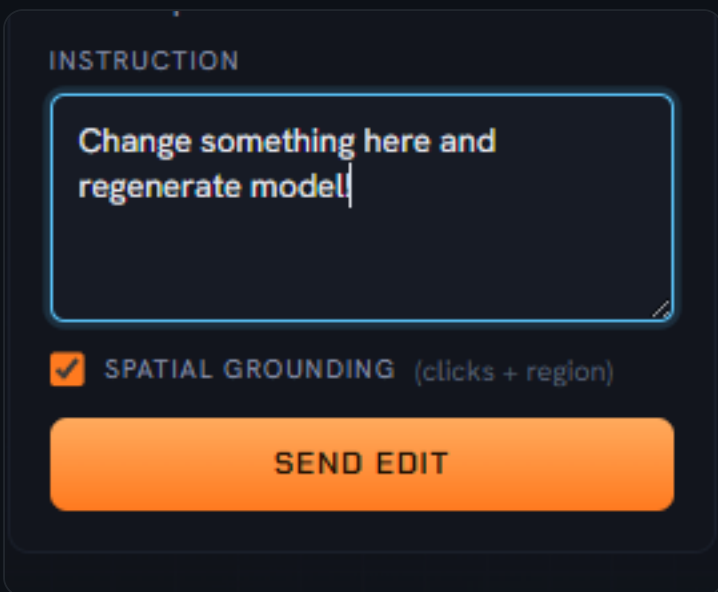
What's done

A complete workflow — from generation to dataset



01 Region selection

With a click, we place several points on the model — selecting regions to edit. Each point is shown by its coordinates, and any of them can be removed.



02 Text editing

We write the instruction in plain language. Spatial grounding mode adds the coordinates and region to it — versus a plain "flat" prompt (for comparison in the research study).



03 History and dataset

Every generation and edit is a separate version: you can load it back into the viewer, rate it 1–5, and export the entire set as a dataset (JSON / CSV).

Stack: React · Three.js · Node/Express · Claude API · Meshy AI

Runs locally without keys (mock mode) — a full demo at no cost.